
How the Two Models Connect

The economic channels of the OG-Core $\times$ CLEWS soft link —
and how each is represented in the model

numbers: the committed M=8 OG-PHL golden record · `ogclews-link` · July 6, 2026



The coupling problem

Two models, complementary blind spots

CLEWS / OSeMOSYS

a linear program: choose capacity & dispatch to meet energy demand at least discounted system cost

- **rich:** technology, sub-annual dispatch, emissions, land & water
- **blind:** demand is *fixed* — no price response, no households, no government budget, no distribution

Each is detailed exactly where the other is blank — so what usefully crosses the seam differs by direction:

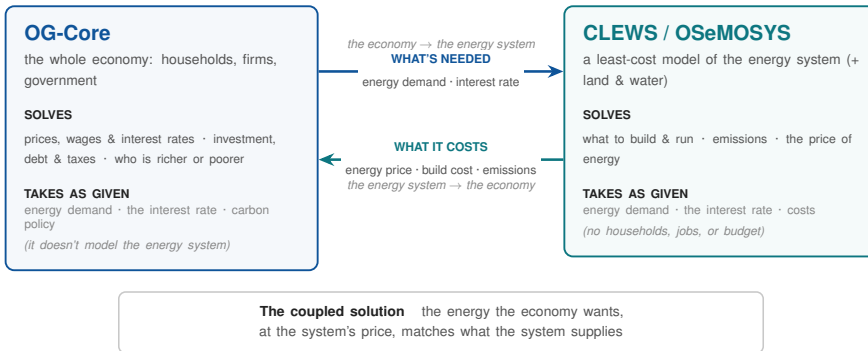
quantities flow from the economy, **prices and physical outcomes** flow from the energy system.

OG-Core

overlapping-generations general equilibrium: households by age $S \times$ lifetime income J , M industries, a government that closes its budget each period

- **rich:** behaviour, saving, fiscal policy, incidence, demography
- **blind:** no energy technology — energy is not a priced input to production

The structural seam



Each model is detailed exactly where the other is blank.

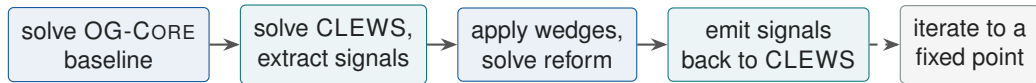
Quantities flow right (energy demand, the interest rate); prices and physical outcomes flow left (the energy price, build cost, emissions). The models stay separate and pass a few auditable signals.

Why a *staged soft* link, not one big model

A hard link is ruled out structurally, not practically:

- OG-CORE solves an OLG *transition path* period by period; OSEMOSYS is one perfect-foresight LP over all years at once
- OG-CORE exposes no scalar welfare into which the LP could be folded; no shock can be injected mid-solve

So the models stay separate and the link runs in stages:



Validated today: the single pass (solid). The iterated loop (dashed) is designed — the emitters already produce well-formed CLEWS inputs — but not yet wired.

Signals, wedges, and the load-bearing dual

The vocabulary: **signal** → **wedge** → **channel**

Signal a quantity carried out of a *solved* CLEWS scenario: the dual on a commodity balance (a marginal cost), the capex a build-out requires, emissions, the dual on an emissions cap (an implied carbon price).

The vocabulary: signal → wedge → channel

Signal a quantity carried out of a *solved* CLEWS scenario: the dual on a commodity balance (a marginal cost), the capex a build-out requires, emissions, the dual on an emissions cap (an implied carbon price).

Wedge the OG-CORE parameter that expresses it: the consumption-tax vector τ^c , an industry productivity Z_m , the public-investment share α_I , mortality $\rho(s, t)$, effective labour $e(t, s, j)$. (“Wedge” in the public-finance sense — the simplest channel literally inserts a tax wedge between producer and consumer price.)

The vocabulary: signal → wedge → channel

Signal a quantity carried out of a *solved* CLEWS scenario: the dual on a commodity balance (a marginal cost), the capex a build-out requires, emissions, the dual on an emissions cap (an implied carbon price).

Wedge the OG-CORE parameter that expresses it: the consumption-tax vector τ^c , an industry productivity Z_m , the public-investment share α_I , mortality $\rho(s, t)$, effective labour $e(t, s, j)$. (“Wedge” in the public-finance sense — the simplest channel literally inserts a tax wedge between producer and consumer price.)

Channel a named, guard-railed map from one signal to one wedge — plus the validation that fires when the mapping is large or ill-posed.

In code: plain functions in `channels.py` (OG-CORE→CLEWS ones carry `emit_` and run after the reform solves). Country-agnostic: industry indices and counts are *discovered* per run from the country’s own calibration, never hard-coded.

The load-bearing signal: the LP dual

In the energy program every commodity carries a balance constraint — supply must meet demand in every timeslice and year. Its **dual** is the marginal system cost of delivering one more unit; under the competitive reading, *the price that would clear that quantity*.

1. It closes an elasticity gap

CLEWS demand is price-*inelastic* by construction. Handing the dual to OG-CORE households — Stone–Geary demand, unit-elastic above a subsistence floor — supplies the demand response the optimiser lacks.

2. It reveals incidence

A single “cost” is silent on who bears it. $S \times J$ households distribute the burden; because energy is a partial necessity, an energy-price change is *regressive* — invisible to a single-agent core.

Soft-link equilibrium (the target)

$D^* = D_{\text{OG}}(\pi^*)$ and $\pi^* = \pi_{\text{C}}(D^*)$ — one pass evaluates π_{C} at *baseline* demand; iterating to the fixed point is loop closure

Three doors for one energy cost

OG-CORE production has no energy input — an energy-cost signal must enter through one of three routes:

- A demand-side tax wedge** — $1 + \tau_{t,i_e}^{c,r} = g_t (1 + \tau_{t,i_e}^{c,b})$, g_t the reform/base price ratio. Reuses existing structure, delivers incidence; but household-only, and the “revenue” is phantom. Empirically $\approx$ output-neutral.
- B supply-side productivity haircut, I–O calibrated** — cost-push $\hat{c} = (I - A^T)^{-1} \theta \hat{p}_E$ applied as per-industry Z_m haircuts. Right sign, materially contractionary; but firms cannot substitute away and the magnitude is a proxy.
- C energy as a priced production input** — a CES nest of value added and purchased energy; dual-consistent, closes the loop on quantities. The structural ideal; requires a core extension (unbuilt).

The route is a first-order modelling choice

Routes A and B carry the *same* physical cost to opposite macro verdicts — roughly neutral vs. sharply contractionary. That is why the dual-consistent representation is the target, not a refinement.



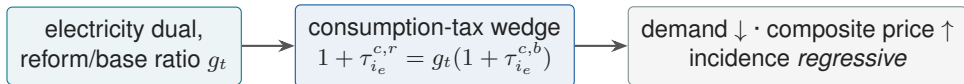
The eight channels

The map

Channel	Dir.	CLEWS signal	OG-CORE object	First-order effect
energy_price	c→o	energy-balance dual / cost ratio	τ^c on the energy good	demand ↓; regressive incidence
carbon	pol	carbon price (\$/tCO ₂)	τ^c add-on + CLEWS emissions penalty	demand ↓; emissions ↓
investment	c→o	public grid/T&D capex	$\alpha_I \rightarrow K_g$	public capital ↑; crowding-out, debt
capital_intensity	c→o	power-fleet capital-cost share	γ_m (energy capital exponent)	energy price ↓; energy $K \downarrow$
energy_capex	pol	ITC rate (policy response)	cost of capital (inv_tax_credit)	capital reallocates into energy
health	c→o	PM _{2.5} emissions ratio	$\rho(s, t), e(t, s, j)$	mortality ≈ 0 ; morbidity ↑ labour
discount_rate	o→c	—	reads $r_p \rightarrow$ DiscountRate	shifts the build mix (via the loop)
demand	o→c	—	reads $Y_m/C_i \rightarrow$ demand	more capacity (via the loop)

four CLEWS→OG-CORE · two policy levers applied once to both sides · two OG-CORE→CLEWS

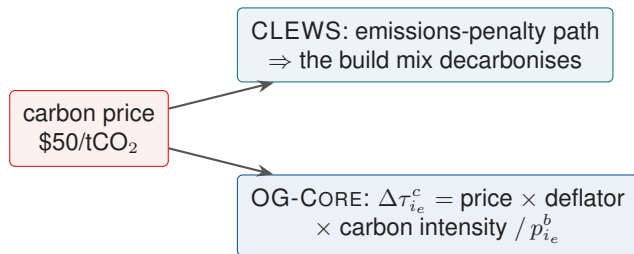
energy_price — the dual reaches household budgets



- households face $(1 + \tau_{i_e}^c) p_{i_e}$; energy quantity responds — the elasticity CLEWS lacks
- with a subsistence floor ($c_{\min} > 0$), poorer households spend a larger energy share $\Rightarrow$ they bear more of a price change, even when the revenue is recycled (clean_incidence: $C_{ss} = -0.087\%$ under full recycling)
- the price is the country's *real* CLEWS price (“auto”: curated cost-of-electricity index, else the OSEMOSYS commodity dual) — for the Philippine pair it is **near-flat**, so the standalone effect is $Y_{ss} = -0.0005\%$: honestly transmitted, the data does little
- the earlier $\pm 20\%$ stress runs proved the *transmission*; they are retained only for the route comparison

maturity: Route A on the real dual; the coupled run transmits the same price through the composite (next section).

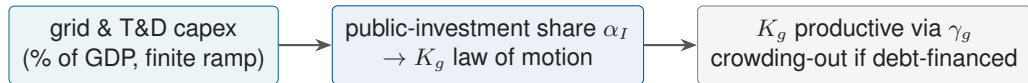
carbon — one price, two sides, counted once



- one exogenous price enters *both* models along the same path — never inferred a second time from a CLEWS shadow price
- OG-CORE firms buy no energy, so only *household* energy carbon is priced on the macro side; industrial carbon is priced in CLEWS
- revenue is recyclable (transfers); standalone it is the clearest contraction: $Y_{ss} = -0.040\%$

the *level* rides on the uncalibrated unit/deflator bridge — mechanics validated, magnitude illustrative.

investment — public grid capital



- this is where crowding-out *legitimately* lives: debt-financed spending competes for the savings pool, raises r , displaces private investment
- public-grid capital and private-generation capital touch different objects — coupling both is complementary, not double counting
- Philippine scenario: the *public* grid-capex delta is ≈ 0 (the transition is generation-side) $\Rightarrow$ the channel is inert here ($Y_{ss} = +0.000\%$); its mechanics are validated and scale with a non-zero delta

The capital pair I: capital_intensity — a worked example

Signal: reform/base ratio of the power fleet's capital-cost share $\Rightarrow$ scales the energy industry's capital exponent γ_m (here +12.2%).

The model's own first-order condition (Cobb–Douglas, $\varepsilon = 1$): the user cost ρ_m and taxes cancel in the reform/base ratio, leaving an exact identity —

$$K_m = \frac{\gamma_m p_m Y_m}{\rho_m} \quad \Rightarrow \quad \hat{K}_E = \hat{\gamma}_E \cdot \hat{p}_E \cdot \hat{Y}_E$$

	$\hat{\gamma}_E$	$\hat{p}_E$	$\hat{Y}_E$	$\hat{K}_E$ (obs.)	identity
energy industry ($m_e = 2$)	+12.2%	−42.8%	+20.9%	−22.4%	−22.4%

- the higher exponent cuts the sector's unit cost $\rightarrow$ under zero profit the electricity price **collapses** −42.8%
- demand is elastic enough that output **rises** +20.9% — but revenue $p_m Y_m$ shrinks by more than $\hat{\gamma}$ and $\hat{Y}$ add back $\Rightarrow$ capital **falls** −22.4% (identity closes to 0.03%)

Raising the capital *share* sheds capital: this is a factor-share and energy-price lever, *not* a capital draw-in.

The capital pair II: energy_capex — the actual draw-in lever

The capital-*attraction* force is a structurally different instrument: an **investment tax credit** on the energy industry.

$$\rho_m = \frac{r + \delta - \tau_m^b \delta \tau_m - \tau_m^{\text{inv}} \delta}{1 - \tau_m^b}$$

the ITC enters the *user-cost denominator*, where γ_m is absent $\Rightarrow$ capital demand shifts *out* at given r

γ_m (capital_intensity)

moves factor *demand* through the production exponent; absent from ρ_m . Result: cheaper electricity, *less* energy capital.

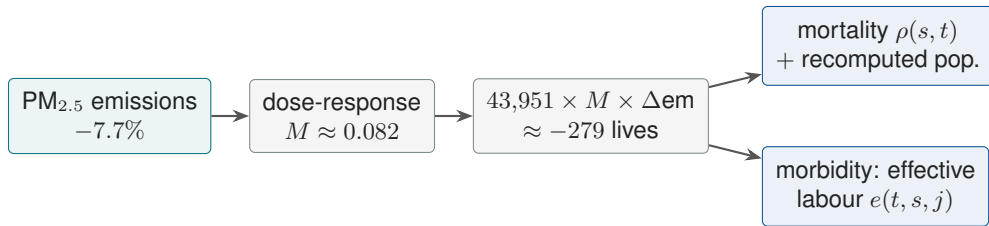
ITC (energy_capex)

moves the *cost of capital*; absent from the exponents. Result: capital reallocates *into* energy. (Defined channel; not run standalone in the present battery.)

The lesson

Two levers, two different objects — *not* two views of one capex. The model's own FOC exposes the mislabeling that partial-equilibrium intuition (“capital-intensive $\Rightarrow$ pulls capital in”) would smuggle into the results; the no-double-counting discipline forbids stacking them.

health — emissions to demography and labour



- real inputs: the IHME GBD ambient-PM_{2.5} burden (43,951 attributable deaths, PHL 2023) with real age profiles; M = energy mass-share $\times$ concentration-response elasticity — the $\times M$ is what separates -279 from the naive -3,400
- **mortality**: PM_{2.5} deaths skew elderly $\rightarrow$ saved lives are mostly retirees $\rightarrow$ long-run output ≈ 0
- **morbidity**: working-age disability falls $\rightarrow$ effective labour rises — *the* macro gain (+0.19% near-term; $Y_{ss} = -0.003\%$)

The signature: a small negative steady state, a larger positive transition — endogenous demography from an emissions signal, invisible to a static core. Remaining assumptions: linearity of the mapping, the morbidity elasticity.

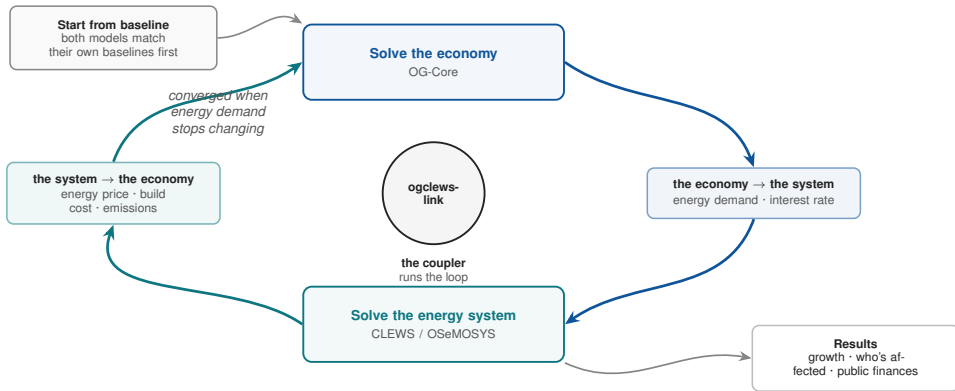
The back-channels: `discount_rate` and demand

Both run *after* the reform solves (`emit_ prefix`) and write well-formed CLEWS inputs:

- `discount_rate`: reads the equilibrium market return r_p (a multi-period mean) $\rightarrow$ CLEWS `DiscountRate`. Energy planning at the *model-consistent* cost of capital: a lower r_p favours capital-heavy, long-lived assets.
- `demand`: reads a reform/base activity ratio (Y_m or C_i , raised to an elasticity) $\rightarrow$ scales CLEWS annual demand. In a single pass the ratio is ≈ 1 — latent by construction.

They move the macro economy only once the loop iterates — the quantity link of the MESSAGE-MACRO/TIMES-MACRO tradition, but feeding an OLG core instead of a representative agent.

Closing the loop



Re-solve the two models in turn until energy demand stops changing. Validated today: the single pass; naive price feedback can oscillate, so the loop needs damping.



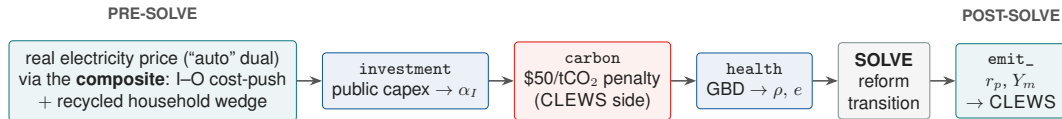
Counting each signal once

Five principles, each enforced by a guard

1. **One carbon price.** It enters once — a CLEWS penalty and an OG-CORE wedge on the same path — never *also* inferred from a CLEWS shadow price.
2. **One view of an energy-cost shock.** A τ^c wedge, a carbon τ^c , and a Z_m haircut are alternative doors for the *same* cost — never stacked.
3. **One view of generation capital.** Cost-push *or* an ITC — and γ_m is *not* a third interchangeable view: it is a factor-income object that sheds capital (the worked example).
4. **Revenue must be routed.** A τ^c wedge mechanically raises consumption-tax revenue; recycle it (transfers, public investment) or finance it explicitly — else a hidden fiscal expansion rides along.
5. **Public and private capital are distinct.** $\alpha_I \rightarrow K_g$ and the firm's own K touch different objects; coupling both is complementary.

Does it behave? The Philippine coupled run

Anatomy of the flagship coupled run

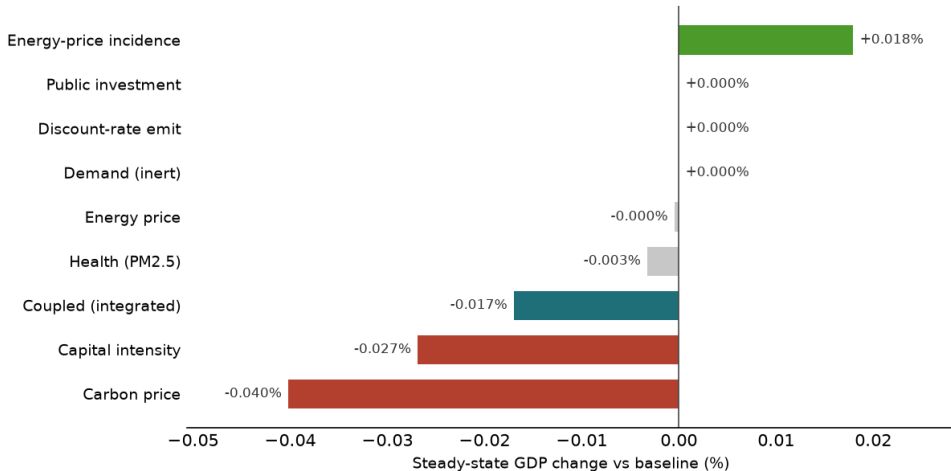


- the Philippine electricity price is near-flat $\Rightarrow$ the energy channels contribute almost nothing (electricity output moves -0.18%); **the macro response is carried by health**
- the carbon price rides the CLEWS side here (one-price principle); the -0.040% on the previous slides is its *standalone* OG-CORE wedge

% change from baseline	steady state	t_0	t_{10}
output Y	-0.017	$+0.165$	$+0.395$
consumption C	$+0.008$	$+0.195$	$+0.467$

PHL OG×CLEWS — steady-state macro impact by channel

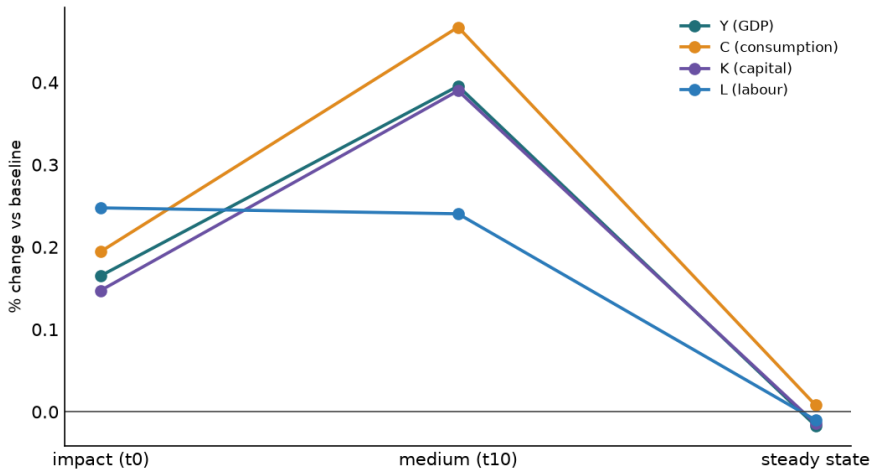
M=8 · ogcore 0.16.3 · real CLEWS 'auto' electricity price (no illustrative +20%)



Per-channel steady-state output impact, standalone reforms on the M=8 Philippine calibration. `energy_price` ≈ 0 on the near-flat real dual; carbon -0.040% is the clearest contraction; `capital_intensity` -0.027% aggregate (but reshapes the energy sector); health -0.003% at steady state yet $+0.19\%$ near-term.

PHL coupled result over the transition

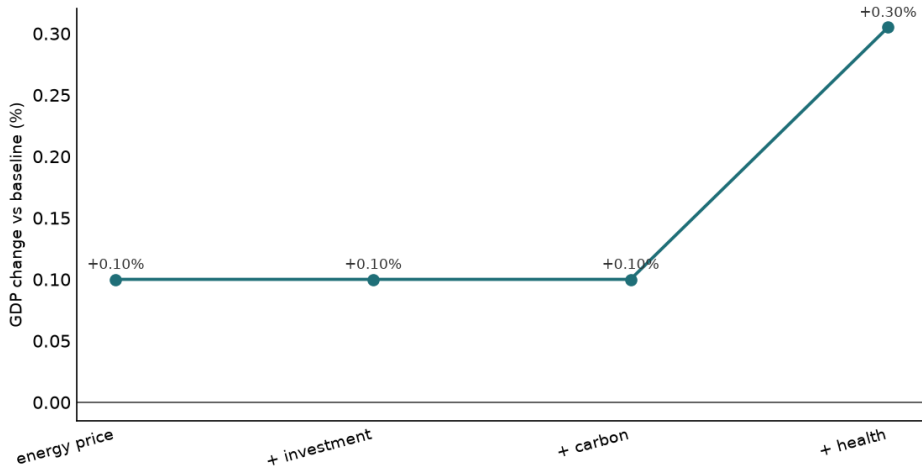
real CLEWS energy + public investment + carbon + GBD health · M=8 · ogcore 0.16.3



The coupled transition: steady state -0.017% , but $+0.165\%$ at t_0 rising to $+0.395\%$ by t_{10} — the working-age morbidity gain along the path, added retirees in the long run. The horizon split is the health channel's signature, and it is invisible to a static core.

Cumulative channel contributions to the real coupled result

each step adds one channel; top step = coupled · M=8 · real CLEWS energy



Cumulative decomposition: channels layered so the top layer *equals* the coupled run — every effect attributable to a named mechanism. That auditability is the point of the channel architecture.

Status

Where the coupling stands

Validated today

- eight channels, each one signal → one wedge, guard-railed
- single CLEWS→OG-CORE pass on *real* inputs: the real electricity dual, the real GBD burden
- cross-environment solve (the link orchestrates; the country's OG-CORE runs in its own interpreter)
- regression-locked golden record; signs and mechanics as theory predicts

Open, in order

- **loop closure** — the emitters are ready; the outer re-solve-and-iterate is not wired
- **the unit/deflator bridge** — carbon and investment *levels* are illustrative until it is calibrated
- **Route C** — energy as a priced production input: the structural destination

One line: A small, auditable library of economic mechanisms — each signal counted once, each effect attributable — rather than one opaque coupling.